

MEIRINGEN, Switzerland

WMO: 066140

Lat: 46.743N

Lon: 8.110E

Elev: 1900

StdP: 13.71

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.2	(c) 16.1	(d) 8.2	(e) 9.0	(f) 13.0	(g) 12.3	(h) 11.0	(i) 19.3	(j) 22.7	(k) 35.1	(l) 18.3	(m) 38.4	(n) 4.7	(o) 100	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	83.8	67.9	80.2	66.0	76.9	64.3	68.9	80.3	67.1	77.5	65.5	74.9	6.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	97.9	74.8	62.9	92.1	72.4	61.5	87.4	70.4	34.2	80.0	32.6	77.6	31.3	74.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.9	12.1	5.5	89.0	4.7	2.7	2.1	90.9	-0.6	92.5	-3.3	94.0	-6.7	95.9
			5.1	72.6	4.4	2.6	2.0	74.5	-0.6	76.0	-3.0	77.5	-6.2	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	30.8	32.7	40.7	48.8	54.4	62.4	65.6	63.9	57.5	50.4	39.4	33.5
	DBStd	13.79	7.20	7.30	6.64	6.90	6.80	6.19	5.41	5.86	5.36	6.09	7.40	6.49
	HDD50	2427	596	485	293	102	36	1	0	0	5	69	326	514
	HDD65	6236	1061	904	754	485	334	120	56	91	232	453	768	978
	CDD50	1852	0	0	4	67	171	374	484	433	229	81	8	1
	CDD65	188	0	0	0	0	4	43	75	59	7	0	0	0
	CDH74	1334	0	0	0	4	49	339	574	347	21	0	0	0
	CDH80	289	0	0	0	0	6	72	141	70	0	0	0	0
	WSAvg	4.1	3.6	4.4	5.3	5.1	4.5	4.4	4.3	3.7	3.7	3.2	3.6	3.3
	PrecAvg	54.8	3.5	3.7	3.6	3.9	5.6	5.7	5.9	5.9	4.4	3.9	4.5	4.2
(15) Precipitation	PrecMax	71.9	9.6	11.7	7.2	7.9	9.9	9.1	11.3	12.2	8.0	7.2	13.6	12.1
	PrecMin	39.8	0.3	0.8	0.9	1.0	1.5	3.0	2.9	1.6	2.0	0.1	0.6	0.0
	PrecStd	6.4	2.3	2.5	1.9	1.5	2.5	1.5	1.9	2.1	1.5	1.8	2.7	2.6
	PrecStd	6.4	2.3	2.5	1.9	1.5	2.5	1.5	1.9	2.1	1.5	1.8	2.7	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	55.1	64.1	74.2	78.8	86.2	88.1	86.0	76.6	69.1	63.0	53.4
		MCWB	40.8	41.7	49.5	56.5	62.6	68.2	69.5	68.7	61.9	54.0	50.0	43.3
	2%	DB	44.6	49.0	59.4	70.0	73.8	82.0	84.2	81.7	73.3	64.8	57.1	48.7
		MCWB	39.7	40.8	47.2	54.8	59.9	66.1	67.7	68.0	61.5	53.3	47.6	41.5
	5%	DB	41.7	45.8	56.4	66.1	70.8	77.4	80.4	77.4	69.8	62.0	53.5	45.3
		MCWB	38.4	38.5	45.7	52.3	58.3	64.5	65.9	65.4	59.9	53.0	46.5	39.7
	10%	DB	39.4	42.7	52.1	62.2	67.1	74.3	77.0	74.4	66.7	59.3	50.2	42.6
		MCWB	36.9	37.7	43.7	50.0	57.0	62.9	64.7	64.1	58.7	52.6	45.2	38.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	43.9	50.3	58.3	64.1	70.1	71.9	71.0	65.3	59.9	51.7	45.2
		MCDB	45.3	50.6	60.7	71.8	75.8	82.2	86.3	82.9	74.0	64.0	58.9	48.8
	2%	WB	40.5	41.4	48.4	55.7	61.6	67.5	68.9	68.8	63.0	57.5	49.5	42.9
		MCDB	43.1	47.1	58.8	68.6	72.4	79.0	80.7	79.7	70.4	61.2	55.7	47.3
	5%	WB	38.6	39.5	46.5	53.3	59.2	65.6	67.0	66.9	61.2	55.3	47.9	41.2
		MCDB	41.0	44.1	55.1	63.7	68.3	76.1	77.7	75.9	68.2	58.7	52.5	44.6
	10%	WB	37.1	37.9	44.5	51.0	57.3	63.7	65.4	65.0	59.4	53.6	45.9	39.1
		MCDB	39.1	41.8	50.8	60.5	65.7	73.1	75.2	73.2	65.7	57.9	49.3	41.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	13.2	16.3	18.5	17.7	17.7	18.4	16.9	15.4	13.8	11.2	11.2
		MCDBR	13.2	18.6	23.9	27.3	25.4	23.5	24.7	21.9	19.9	18.4	16.2	15.5
		MCWBR	8.7	11.3	13.8	14.5	13.7	12.9	12.3	11.8	11.6	10.3	9.1	9.7
	5% WB	MCDBR	11.0	16.9	22.4	25.1	22.8	22.0	23.1	21.6	18.0	14.4	15.0	12.5
		MCWBR	8.0	10.7	13.4	14.1	12.9	12.4	12.4	11.9	11.0	9.3	9.5	8.5
(40) Clear-Sky Solar Irradiance	taub		0.288	0.321	0.376	0.428	0.450	0.454	0.455	0.439	0.405	0.369	0.322	0.285
	taud		2.299	2.197	2.100	2.049	2.076	2.131	2.133	2.188	2.261	2.324	2.356	2.323
	Ebn at Noon		253	265	264	261	260	259	257	256	254	245	237	239
	Edh at Noon		24	33	43	49	50	48	47	43	36	29	23	21
(44) All-Sky Solar Radiation	RadAvg		451	740	1117	1398	1552	1825	1799	1564	1213	790	462	372
	RadStd		54	96	160	213	149	167	157	144	115	87	64	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (119 neighbors)	N/A	N/A	N/A	+1.31	+0.74	+0.58	N/A	N/A	+104	+29

Nomenclature: See separate page